Use Case Diagram
University Library System

1. Objective

To understand and practice designing a UML Use Case Diagram by identifying actors, use cases, system boundaries, and relationships.

2. Learning Outcomes

Identify actors and use cases from a system description.
Draw a UML Use Case Diagram with a system boundary.
Represent associations and use <<include>> and <<extend>> relationships correctly.

3. Tools Required

Draw.io (diagrams.net), StarUML, another UML modeling tool, or pen and paper.

4. Problem Description

A university wants to develop a University Library System to manage library operations. Students can search for books, borrow books, and return them. Librarians manage book records and check overdue fines. The system automatically sends reminders for overdue books.

5. Lab Tasks

Task 1: Identify Actors. Identify the external users or entities interacting with the system.

Task 2: Identify Use Cases. Identify at least five use cases from the description.

Task 3: Draw the Diagram. Show actors (stick figures), use cases (ovals), the system boundary (rectangle), associations (solid lines), and at least one <<include>> and one <<extend>> relationship (dashed arrows).

Task 4: Explain Relationships. Briefly explain the <<include>> and <<extend>> relationships chosen.

6. Procedure

1. Read the problem description and identify the actors.
2. List the main use cases.
3. Draw a rectangle labeled “University Library System” and place use cases inside it.
4. Place actors outside the rectangle and connect them to use cases.
5. Add appropriate <<include>> and <<extend>> relationships using dashed arrows.
6. Review the diagram for correctness and completeness.

7. Submission Requirements

- List of identified actors and use cases.
- Brief explanation of the include and extend relationships.
- Completed UML Use Case Diagram.

8. Self-assessment Criteria

Criteria	**Marks**
Correct identification of actors	20
At least five use cases	20
System boundary and associations	20
Correct <<include>> relationship	20
Correct <<extend>> relationship	20
Total	**100**